



# Python Assessment Test

30 questions · Multiple choice + coding · Closed-book

QUESTIONS

30

DURATION

60 min

TOTAL MARKS

50

MCQ

25 × 1

CODING

5 × 5

CANDIDATE NAME

ROLL / ID

DATE

BATCH / CLASS

INVIGILATOR SIGN

SCORE / 50

## INSTRUCTIONS

- 1 Tick exactly one option for questions 1–25. There is no negative marking.
- 2 Questions 26–30 are coding. Write readable Python. Partial credit is awarded.
- 3 Do not use a phone, notes, or another person. Raise your hand for clarification.
- 4 Use the answer boxes provided. Cross out any work you do not want marked.

## Part A · Language Fundamentals

Variables, types, operators &amp; strings · 1 mark each

1 Which of the following is a valid Python variable name?

☐ A 1var☐ B var\_1☐ C var-1☐ D class

2 What is the output of `x = 5; print(type(x))` ?

☐ A int☐ B <class 'int'>☐ C integer☐ D type int

3 Which built-in function is used to take input from the user?

☐ A scan()☐ B input()☐ C get()☐ D read()

4 What is the output of `10 // 3` ?

A 3.33

B 3

C 4

D 3.0

5 Which operator is used for exponentiation in Python?

A ^

B \*\*

C exp

D ^^

6 What is the result of `5 > 3 and 2 < 1` ?

A True

B False

C Error

D None

7 Given `s = 'Python'` , what does `print(s[0])` display?

A P

B y

C t

D n

8 Which method converts a string to uppercase?

A upper()

B uppercase()

C toUpper()

D caps()

9 What is the value of `len('Hello')` ?

A 4

B 5

C 6

D Error

**Part B · Data Structures**

Lists, tuples, sets &amp; dictionaries · 1 mark each

**10** Lists in Python are:☐ A Immutable☐ B Mutable☐ C Fixed size☐ D None of these**11** If `nums = [1, 2, 3]` then `nums.append(4)` results in:☐ A `[1, 2, 3]`☐ B `[1, 2, 3, 4]`☐ C Error☐ D None**12** The first index of a sequence in Python is:☐ A 1☐ B 0☐ C -1☐ D Depends on the type**13** Tuples in Python are:☐ A Mutable☐ B Immutable☐ C Dynamic☐ D None of these**14** Which syntax correctly creates a tuple?☐ A `[]`☐ B `{}`☐ C `()`☐ D `<>`**15** Sets in Python allow:☐ A Duplicate values☐ B Unique values only☐ C Guaranteed order☐ D Indexing

16 What is the value of {1, 2, 2, 3} ?

A {1, 2, 2, 3}

B {1, 2, 3}

C Error

D [1, 2, 3]

17 A dictionary stores data as:

A Key-value pairs

B Lists

C Tuples

D Sets

18 If d = {'a': 1} , what does print(d['a']) display?

A a

B 1

C Error

D None

## Part C · Control Flow & Functions

Conditionals, loops & function basics · 1 mark each

19 What is the output of if 5 > 3: print('Yes') ?

A Yes

B No

C Error

D Nothing

20 Which keyword is used for multiple conditions after if ?

A else if

B elif

C elseif

D ifelse

21 What does for i in range(3): print(i) print?

A 1 2 3

B 0 1 2

C 0 1 2 3

D 1 2

22 Which loop is condition-based (repeats while a condition is true)?

A for

B while

C do-while

D foreach

23 After `i = 0` then `while i < 3: i += 1`, what is the final value of `i`?

A 2

B 3

C 4

D Error

24 Which keyword is used to define a function in Python?

A function

B def

C fun

D define

25 If `def add(a, b): return a + b` then `add(2, 3)` returns:

A 23

B 5

C Error

D None

## Part D · Coding

Write complete, working Python code · 5 marks each

26 Even or Odd

5 marks

Write a program that reads an integer and prints whether it is even or odd.

Hint: Use the modulo operator `%` and an `if / else` statement.

# write your code here

**27 Sum of first N natural numbers****5 marks**

Write a program that reads N and prints the sum of the first N natural numbers.

*Hint: A loop or the formula  $N * (N + 1) / 2$  is acceptable.*

# write your code here

**28 Reverse a string****5 marks**

Write a program that reads a string and prints it reversed.

*Hint: Slicing  $s[::-1]$  or a loop may be used.*

# write your code here

**29 Maximum in a list****5 marks**

Write a function that takes a list of numbers and returns the largest value.

*Hint: Do not use `max()` unless you also show a loop-based solution.*

# write your code here

## 30 Factorial — loop and recursion

5 marks

Write two functions that compute the factorial of N: one using a loop, and one using recursion.

*Hint: Handle  $0! = 1$  . Show both implementations clearly.*

```
# write your code here
```

## END OF QUESTION PAPER

Check that all 30 questions are attempted before submitting.